

<u>integer quotient</u>		<u>remainder</u>	<u>coefficient</u>
$10/2=5$	+	0	$a_2=0$
$5/2=2$	+	1	$a_3=1$
$2/2=1$	+	0	$a_4=0$
$1/2=0$	+	1	$a_5=1$

answer: $(41)_{10} = (a_5 a_4 a_3 a_2 a_1 a_0)_2 = (101001)_2$

The arithmetic process can be manipulated more conveniently as follows:

<u>integer</u>	<u>remainder</u>
41	1
20	0
10	0
5	0
2	1
1	0
0	1

101001 = answer

The conversion from decimal integers to any base- r system is similar to the above example, except that division is done by r instead of 2.

Example 1-2 Convert decimal 153 to octal. The required base r is 8. First, 153 is divided by 8 to give an integer quotient of 19 and a remainder of 1. Then 19 is divided by 8 to give an integer quotient of 2 and a remainder of 3. Finally, 2 is divided by 8 to give a quotient of 0 and a remainder of 2.

This process can be conveniently manipulated as follows:

153	
19	1
2	3
0	2

= $(231)_8$

The conversion of a decimal fraction to binary is accomplished by a method similar to that used for integers. However, multiplication is used instead of division, and integers are accumulated instead of remainders.

Example 1-3: Convert $(0.6875)_{10}$ to binary. First, 0.6875 is multiplied by 2 to give an integer and a fraction. The new fraction is multiplied by 2 to give a new integer and a new fraction. This process is continued until the fraction becomes 0 or until the number of digits have sufficient accuracy.